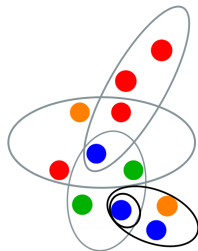
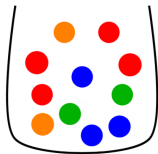


From urns to hypergraphs: a Pólya model

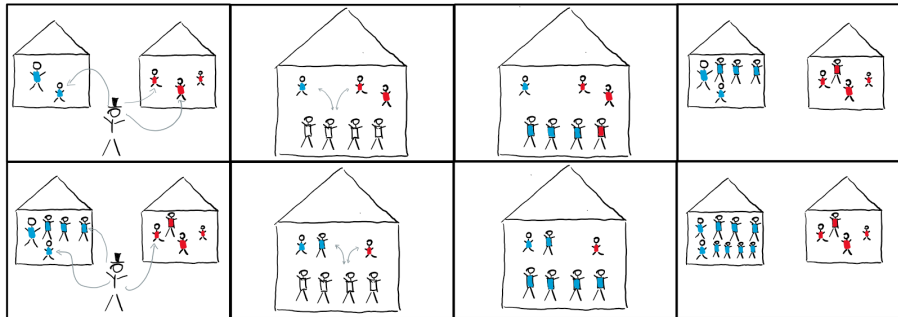
Natalia Jeszka
University of Twente

based on work in progress with
Clara Stegehuis and Martijn Gösgens



Discrete Random Structures II
Będlewo, 2026

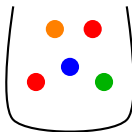
The cartoon



Classical Pólya urn

- balls of d colors
- urn-state at time $t \geq 0$

$$\mathcal{U}(t) = (\mathcal{U}_1(t), \dots, \mathcal{U}_d(t)).$$

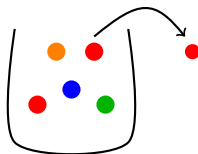


Classical Pólya urn

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$$\mathcal{U}(t) = (\mathcal{U}_1(t), \dots, \mathcal{U}_d(t)).$$

- at every time step:
 - ▶ draw one ball uniformly at random (outcome color: j)



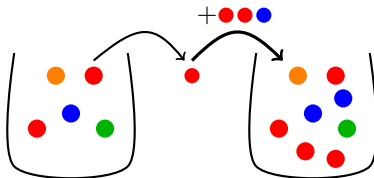
Classical Pólya urn

- balls of d colors
- urn-state at time $t \geq 0$

$$\mathcal{U}(t) = (\mathcal{U}_1(t), \dots, \mathcal{U}_d(t)).$$

- at every time step:
 - ▶ draw one ball uniformly at random (outcome color: j)
 - ▶ replace it according to the rule $M = (M_{ij})_{1 \leq i, j \leq d}$:

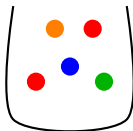
$$\mathcal{U}(t+1) = (\mathcal{U}_1(t) + M_{1j}, \dots, \mathcal{U}_d(t) + M_{dj}).$$



Multi-drawing Pólya urn

- balls of d colors
- urn-state at time $t \geq 0$

$$\mathcal{U}(t) = (\mathcal{U}_1(t), \dots, \mathcal{U}_d(t)).$$

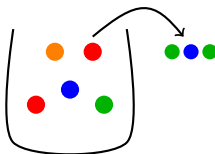


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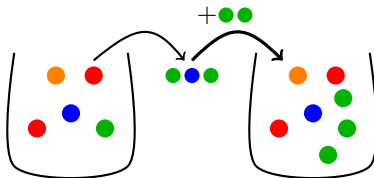


Multi-drawing Pólya urn

- balls of d colors
- urn-state at time $t \geq 0$

$$\mathcal{U}(t) = (\mathcal{U}_1(t), \dots, \mathcal{U}_d(t)).$$

- at every time step:
 - ▶ draw m balls uniformly at random (with replacement)
 - ▶ replace them according to the rule



Replacement rule

A *replacement tensor* is a collection

$$R = (R_{i,j_1,\dots,j_m})_{1 \leq i,j_1,\dots,j_m \leq d},$$

where $R_{i,j_1,\dots,j_m}$ denotes the number of balls of color i added after drawing the multiset of colors $(j_1, \dots, j_m)$:

$$\mathcal{U}(t+1) = (\mathcal{U}_1(t) + R_{1,j_1,\dots,j_m}, \dots, \mathcal{U}_d(t) + R_{d,j_1,\dots,j_m}).$$

Example: 2 colors, 2 draws

$$R = \left(\begin{array}{cc|cc} 3 & 1 & 0 & 5 \\ 4 & 2 & 2 & 3 \end{array} \right)$$

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Example: 3 colors, 2 draws

$$R = \left(\begin{array}{ccc|ccc|ccc} 2 & 1 & 0 & 1 & 3 & 2 & 0 & 4 & 1 \\ 4 & 2 & 1 & 0 & 2 & 4 & 2 & 1 & 3 \\ 3 & 0 & 5 & 5 & 1 & 1 & 1 & 5 & 2 \end{array} \right)$$

Replacement rule

A *replacement tensor* is a collection

$$R = (R_{i,j_1,\dots,j_m})_{1 \leq i,j_1,\dots,j_m \leq d},$$

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$$\mathcal{U}(t+1) = (\mathcal{U}_1(t) + R_{1,j_1,\dots,j_m}, \dots, \mathcal{U}_d(t) + R_{d,j_1,\dots,j_m}).$$

Example: 2 colors, 3 draws

$$R = \left(\begin{array}{cc|cc||cc|cc} 2 & 0 & 4 & 1 & 4 & 1 & 1 & 5 \\ 1 & 3 & 0 & 2 & 0 & 2 & 2 & 1 \end{array} \right)$$

Replacement rule

A *replacement tensor* is a collection

$$R = (R_{i,j_1,\dots,j_m})_{1 \leq i,j_1,\dots,j_m \leq d},$$

where $R_{i,j_1,\dots,j_m}$ denotes the number of balls of color i added after drawing the multiset of colors $(j_1, \dots, j_m)$:

$$\mathcal{U}(t+1) = (\mathcal{U}_1(t) + R_{1,j_1,\dots,j_m}, \dots, \mathcal{U}_d(t) + R_{d,j_1,\dots,j_m}).$$

From now on, **d colors, 2 draws** case:

$$R = \left(\begin{array}{ccc|ccc|ccc} R_{1,1,1} & \cdots & R_{d,1,1} & R_{1,2,1} & \cdots & R_{d,2,1} & \cdots & R_{1,d,1} & \cdots & R_{d,d,1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ R_{1,1,d} & \cdots & R_{d,1,d} & R_{1,2,d} & \cdots & R_{d,2,d} & \cdots & R_{1,d,d} & \cdots & R_{d,d,d} \end{array} \right),$$

$$\mathcal{U}(t+1) = (\mathcal{U}_1(t) + R_{1,j_1,j_2}, \dots, \mathcal{U}_d(t) + R_{d,j_1,j_2}).$$

Replacement rule as a bilinear map

$$R = \left(\begin{array}{ccc|ccc|ccc} R_{1,1,1} & \cdots & R_{d,1,1} & R_{1,2,1} & \cdots & R_{d,2,1} & \cdots & R_{1,d,1} & \cdots & R_{d,d,1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ R_{1,1,d} & \cdots & R_{d,1,d} & R_{1,2,d} & \cdots & R_{d,2,d} & \cdots & R_{1,d,d} & \cdots & R_{d,d,d} \end{array} \right)$$

Tensor R can be viewed as a bilinear map $\mathcal{R} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ as follows.

For $x, y \in \mathbb{R}^d$, $k \in [d]$ we set

$$\mathcal{R}_k(x, y) = \sum_{i=1}^d \sum_{j=1}^d R_{k,i,j} x_i y_j$$

and $\mathcal{R}(x, y) = (\mathcal{R}_1(x, y), \dots, \mathcal{R}_d(x, y))$.

In particular, for the standard basis vectors e_i, e_j , we have $\mathcal{R}(e_i, e_j) = (R_{1,i,j}, \dots, R_{d,i,j})$.

Replacement rule – assumptions

Tenability (T)

For all $1 \leq i, j, k \leq d$,

$$R_{i,j,k} \geq 0.$$

Balance (B)

There exists $\sigma \in (0, \infty)$ such that, for all $1 \leq j, k \leq d$,

$$\sum_{i=1}^d R_{i,j,k} = \sigma.$$

Ergodicity (E)

There holds

$$\max_{\substack{1 \leq j, j' \leq d \\ 1 \leq k, k' \leq d}} \sum_{i=1}^d |R_{i,j,k} - R_{i,j',k'}| < \sigma.$$

Balls distribution

Theorem (C. Mailler and R. Steiner, '25+)

Let R satisfy assumptions (T), (B), and (E).

Then:

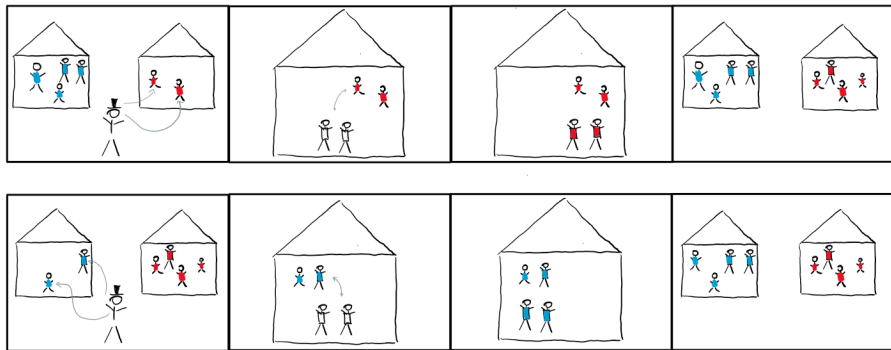
i There exists a unique $x^* \in \{x \in [0, 1]^d : \|x\|_1 = 1\}$ such that

$$\sigma_{x^*} = \mathcal{R}(x^*, x^*).$$

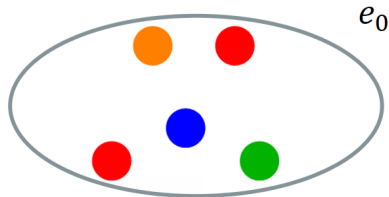
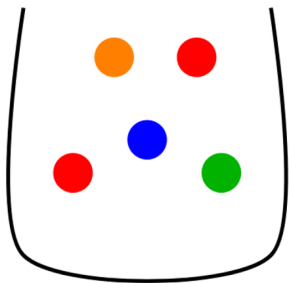
ii The urn-state proportions converge:

$$\frac{\mathcal{U}(t)}{\|\mathcal{U}(t)\|_1} \xrightarrow[\mathbb{P}]{t \rightarrow \infty} x^*.$$

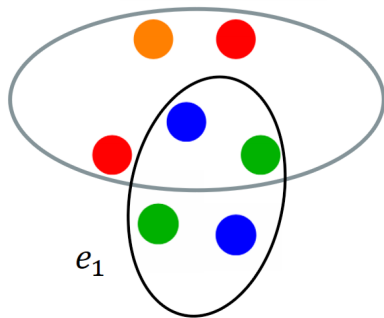
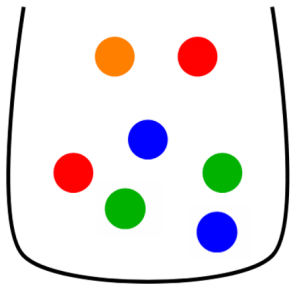
Cartoon – revisited



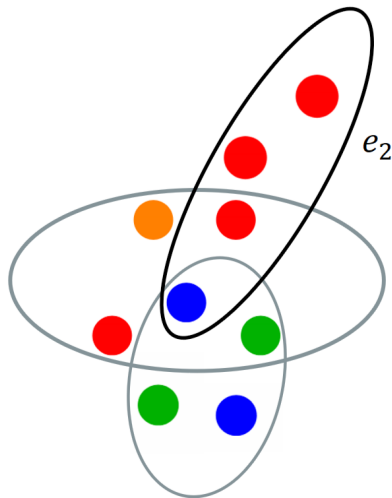
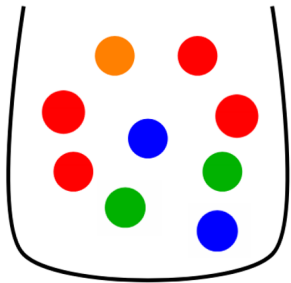
From urns to hypergraphs



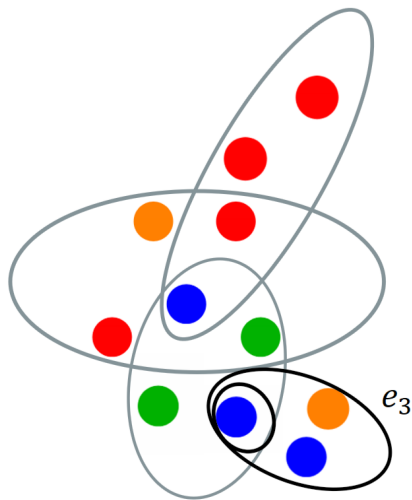
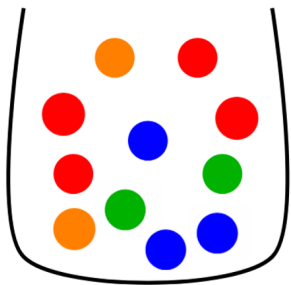
From urns to hypergraphs



From urns to hypergraphs



From urns to hypergraphs



Hyperedge distribution

Fix edge type e .

Let $N_e(t)$ be the number of edges of type e at time t .

Theorem (M. Gösgens, N.J. and C. Stegehuis, '26+)

Let R satisfy assumptions (T), (B), and (E). Then the proportion of edges of type e converges over time to a L^1 -bounded random variable $N_e^{(\infty)}$, i.e.

$$\frac{N_e(t)}{t} \xrightarrow[L^1]{t \rightarrow \infty} N_e^{(\infty)}$$

where $\mathbb{E}[|N_e^{(\infty)}|] < \infty$.

Proof sketch

Decompose

$$M_e(t) = N_e(t) - \sum_{i=0}^{t-1} f_e(\mathcal{U}(i)/\|\mathcal{U}(i)\|_1),$$

where

$$f_e(\mathcal{U}(t)/\|\mathcal{U}(t)\|_1) = \mathbb{E}[\Delta N_e(t) \mid \mathcal{U}(t)/\|\mathcal{U}(t)\|_1].$$

Proof sketch

Normalize

$$\frac{M_e(t)}{t} = \frac{N_e(t)}{t} - \frac{\sum_{i=0}^{t-1} f_e(\mathcal{U}(i)/\|\mathcal{U}(i)\|_1)}{t},$$

where

$$f_e(\mathcal{U}(t)/\|\mathcal{U}(t)\|_1) = \mathbb{E}[\Delta N_e(t) \mid \mathcal{U}(t)/\|\mathcal{U}(t)\|_1].$$

Proof sketch

Take the limit

$$\begin{aligned} \frac{M_e(t)}{t} &= \frac{N_e(t)}{t} - \frac{\sum_{i=0}^{t-1} f_e(\mathcal{U}(i)/\|\mathcal{U}(i)\|_1)}{t}, \\ &\downarrow \\ M_e^{(\infty)}(=0) \end{aligned}$$

where

$$f_e(\mathcal{U}(t)/\|\mathcal{U}(t)\|_1) = \mathbb{E}[\Delta N_e(t) \mid \mathcal{U}(t)/\|\mathcal{U}(t)\|_1].$$

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$$\begin{array}{ccc} \downarrow & & \downarrow \\ 0 & & ? \end{array}$$

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where

$$\begin{aligned} f_e(\mathcal{U}(t)/\|\mathcal{U}(t)\|_1) &= \mathbb{E}[\Delta N_e(t) \mid \mathcal{U}(t)/\|\mathcal{U}(t)\|_1] \\ &= \sum_{\substack{\text{draws potentially} \\ \text{creating } e}} \mathbb{P}[\text{draw} \mid \frac{\mathcal{U}(t)}{\|\mathcal{U}(t)\|_1}] \cdot \mathbb{1}_{\{R_{i,\text{draw}} = \text{nr of color } i \text{ in "e-draw", } i=1,\dots,d\}}. \end{aligned}$$

Proof sketch

Take the limit

$$\begin{array}{ccc} \frac{M_e(t)}{t} & = & \frac{N_e(t)}{t} - \frac{\sum_{i=0}^{t-1} f_e(\mathcal{U}(i)/\|\mathcal{U}(i)\|_1)}{t}, \\ \downarrow & & \downarrow \\ 0 & & f_e(x^*) \end{array}$$

where

$$\begin{aligned} f_e(\mathcal{U}(t)/\|\mathcal{U}(t)\|_1) &= \mathbb{E}[\Delta N_t(e) \mid \mathcal{U}(t)/\|\mathcal{U}(t)\|_1] \\ &= \sum_{\substack{\text{draws potentially} \\ \text{creating } e}} \mathbb{P}[\text{draw} \mid \mathcal{U}(t)] \cdot \mathbb{1}_{\{R_{i,\text{draw}} = \text{nr of color } i \text{ in "e-draw", } i=1,\dots,d\}}. \end{aligned}$$

Randomized model

Randomized model

- $R^{(H)}$ and $R^{(T)}$ – replacement tensors satisfying (T), (B), (E)
- $C_t \sim \text{Bernoulli}(p)$, $p \in [0, 1]$

Let

$$R(C_t) = \mathbb{1}_{\{C_t=H\}} R^{(H)} + \mathbb{1}_{\{C_t=T\}} R^{(T)}.$$

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Let

$$R(C_t) = \mathbb{1}_{\{C_t=H\}} R^{(H)} + \mathbb{1}_{\{C_t=T\}} R^{(T)}.$$

Theorem (M. Gösgens, N.J. and C. Stegehuis, '26+)

Let the urn $\mathcal{U}(t)$ be governed by the replacement tensor $R(C_t)$ and x^ be the unique invariant point of the normalized expected tensor $\mathbb{E}[R(C_t)]/\sigma$, i.e. fulfill*

$$\sigma x^* = \mathbb{E}[\mathcal{R}(C_t)](x^*, x^*).$$

Then the urn-state process converges almost surely to this fixed point

$$\frac{\mathcal{U}(t)}{\|\mathcal{U}(t)\|_1} \xrightarrow[t \rightarrow \infty]{a.s.} x^*.$$

Reference

-  C. Mailler and R. Steiner.
Multi-drawing Pólya urns via labelled random DAGs.
arXiv preprint, 2025.